

xyndrome — Pure-Native Student App (iOS + Android) — Build Plan

Document status: Draft v1 for approval **Date:** 2026-06-13 **Author:** Engineering (Claude Code) **Scope:** Student side only — a 1:1 pure-native rebuild of the existing React/Capacitor LMS student app.

0. Decisions locked (from your answers)

Decision	Choice	Consequence
Technology	Flutter (single Dart codebase)	True native (AOT ARM, no webview). One codebase renders pixel-identical on iOS + Android — best match to "exactly copy the same design + effects."
Billing	External web checkout (PayHere)	Native app opens the existing web payment flow in the system browser. ⚠️ Flagged store-policy risk — see §14.
Offline	Hybrid: online-first app, but offline-first for notes + free content	The app is online-first overall (graceful offline states + the same retry/timeout resilience as the web Axios client), except the user's own notes ("writing own notes") and free (non-premium) study content, which are offline-first — created/edited locally and synced when back online. See §3.3.
Build order	All ~26 screens planned as one build	The plan below is a single end-to-end build (still organized into an ordered task list / WBS in §19 so it's buildable "one task at a time").

What stays identical and is NOT touched: the NestJS v11 backend, the MySQL database, every API endpoint, the design tokens (we mirror their exact values), the logo, and all brand assets. The native apps are a new **frontend only** that talks to the same backend the web app already uses.

1. Goal & non-goals

Goal

Two real native apps (iOS + Android) from **one Flutter codebase** that are a faithful, pixel-accurate replica of the current student web app: same logo, same design language (XYNDROME v2), same color tone, same elements, same effects (glassmorphism, score-ring count-up, flashcard flip, note-canvas + scribble sound, haptics), same screens, same backend, same database. Fully responsive across phone / tablet (iPad) / desktop form factors.

Non-goals (explicitly out of scope)

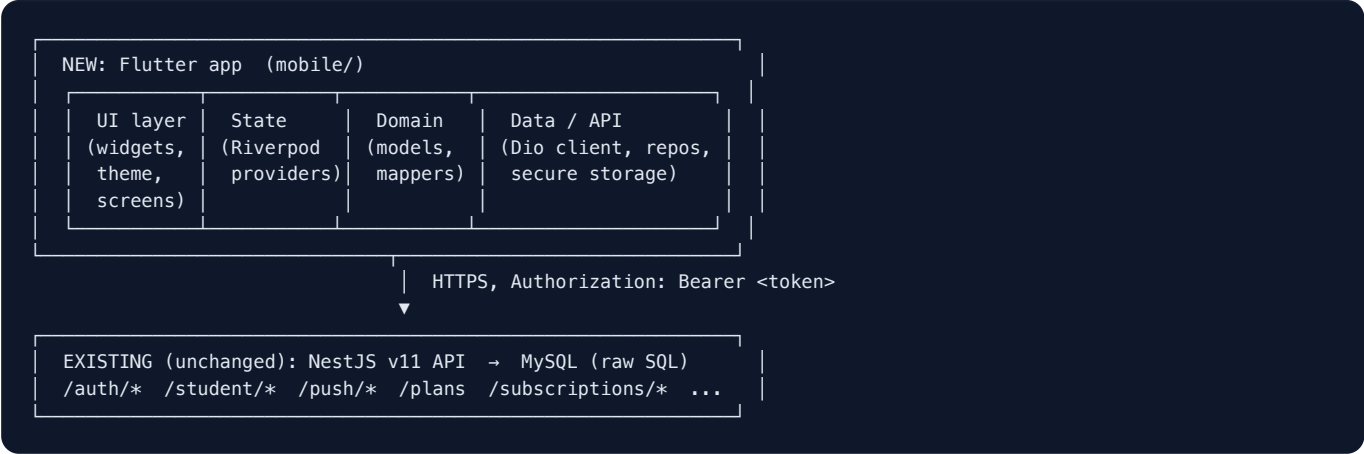
- Admin side** (/admin/*) — student only.
- Backend / DB changes** — reused as-is. (Two small *additive* server touchpoints may be needed; see §9.4 "Open server items.")
- New features** — this is a replication, not a redesign.
- In-app purchase (StoreKit / Play Billing)** — per your choice we use external web checkout for v1.
- Full offline mode for the whole app** — only the notes + free-content slice is offline-first in v1 (§3.3); premium/network features stay online-first.

2. Why Flutter (one paragraph)

The single hardest requirement is "**exactly** the same design and effects on **both** iOS and Android." Flutter draws every pixel itself (Impeller/Skia) rather than mapping to each platform's native widgets, so a glassmorphism card, a conic-gradient score ring, a flashcard flip, and the custom note canvas look **identical** on both platforms from one codebase — no per-platform drift. It compiles

ahead-of-time to native ARM (no webview, unlike Capacitor), gives 60/120 fps animations, and has first-class tools for exactly the things this app needs: `BackdropFilter` (glass), `CustomPainter` (score ring, paper texture, drawing canvas), `AnimationController` (count-up, flip, route transitions), pointer pressure (Apple Pencil / S-Pen), and platform haptics/notifications.

3. High-level architecture



3.1 Where it lives

A new top-level folder in the monorepo: `mobile/` (Flutter project). The existing `frontend/`, `backend/`, etc. are untouched. The Capacitor `frontend/ios` + `frontend/android` projects are left in place (we do not delete them yet — the new apps eventually replace them on the stores).

3.2 Chosen Flutter stack (libraries)

Concern	Web app uses	Flutter equivalent (proposed)	Why
Language	JS (React 19)	Dart 3 + Flutter 3.x (stable)	—
State	Zustand	Riverpod 2	Closest to Zustand's simple, testable global stores; compile-safe.
Routing	React Router 7	go_router	Declarative routes, deep links, redirect guards (auth).
HTTP	Axios	Dio + <code>dio_smart_retry</code>	Interceptors, base-URL, timeout + retry parity with the Axios client.
Secure token	localStorage	flutter_secure_storage	Keychain (iOS) / EncryptedSharedPreferences (Android) — upgrade over web localStorage.
Light cache/prefs	localStorage	shared_preferences + hive	Settings, theme, light response cache (mirrors <code>createTimedApiCache</code>).
SVG	inline JSX SVG	flutter_svg	Render the medical icon set + ported UI icons.
Glass	CSS backdrop-filter	BackdropFilter + saturation shader	See §11.1.
Drawing	DOM canvas	CustomPaint + <code>perfect_freehand</code> (Dart) + pointer pressure	Note canvas / annotations.
Procedural audio	Web Audio API	flutter_soloud (+ platform channel fallback)	Low-latency generated buffers + real-time modulation. See §11.4.
Push	@capacitor/push-notifications	firebase_messaging (FCM + APNs) + flutter_local_notifications	Register token to <code>POST /push/native-token</code> .
Haptics	@capacitor/haptics	HapticFeedback (built-in) + <code>gaussian</code> /platform channel for richer set	Map the semantic haptic vocabulary.
External checkout	web route	url_launcher (external browser)	Per billing decision (§14).
Google sign-in	GIS web	google_sign_in	Posts credential to <code>/auth/google</code> .
Status bar / system UI	@capacitor/status-bar	SystemChrome	Match <code>#0b121b</code> .
Splash	@capacitor/splash	flutter_native_splash	Background <code>#0b121b</code> .
Env config	Vite <code>import.meta.env</code>	<code>--dart-define</code> + a <code>config.dart</code>	Dev vs prod API origin.

All library choices are defaults open to change at kickoff; nothing here forces a specific package beyond the architecture.

3.3 Offline-first slice (notes + free content)

While the app is online-first overall, **the user's own notes and free (non-premium) study content are offline-first** — they must work with no connection and sync opportunistically.

- **Local store:** a structured local DB (**Hive** or **Drift/SQLite**) is the *source of truth* for personal notes and free content; the network is a sync target, not a prerequisite.
- **Scope offline-first:**
 - **Personal notes** — `StudentNotesPage` text + the AI-notes canvas annotations the student authors (create / edit / delete fully offline).
 - **Free content** — lessons / flashcards / quizzes flagged as free (non-subscription) are cached on first view and remain readable offline.

- **Sync model:** local-write-first → background sync queue with last-write-wins + `updatedAt` conflict handling; each pending change carries a dirty flag and retries when connectivity returns (reuses the lesson-annotations endpoints `.../lessons/:lessonId/annotations` and the notes/planner endpoints).
 - **UI:** an "Offline — saved on device, will sync" indicator on notes; free content shows a "Saved for offline" badge; premium/network-only screens show the standard graceful offline state.
 - **Premium content stays online** (and is also a billing/entitlement boundary).
- Open confirmation: exact definition of "free stuff" — assumed = the student's own notes/annotations **plus** any lesson/flashcard/quiz the backend marks as free (non-premium). Flag in §18.

4. Design system port (tokens → Flutter ThemeData)

The web app's tokens live in `frontend/src/shared/styles/00-tokens/` + `01-base/theme.css`. We mirror them **verbatim** into a single Dart source of truth: `mobile/lib/theme/tokens.dart` (raw values) + `app_theme.dart` (light/dark `ThemeData`). The student app default is **dark**.

4.1 Color tokens

Dark mode (student default)

Token	Value
page bg <code>--surface-0</code>	<code>#0A0A0F</code>
recessed <code>--surface-1</code>	<code>#111117</code>
<code>--surface-2</code>	<code>#14141B</code>
<code>--surface-3</code>	<code>#1A1A22</code>
<code>--surface-4</code>	<code>#20202A</code>
card <code>--surface-card</code>	<code>#16181F</code>
raised <code>--surface-card-elevated</code>	<code>#1C1F27</code>
glass	<code>rgba(10,10,15,0.82)</code> / strong <code>0.94</code> / subtle <code>rgba(12,12,18,0.70)</code>
text strong <code>--ink-strong</code>	<code>#F8FAFC</code>
text medium <code>--ink-medium</code>	<code>#CBD5E1</code>
text soft <code>--ink-soft</code>	<code>#94A3B8</code>
text muted <code>--ink-muted</code>	<code>#8294AE</code>
line soft / medium / strong	<code>rgba(203,213,225,0.13 / 0.20 / 0.28)</code>
brand primary	<code>#60A5FA</code> (hover <code>#93C5FD</code> , light <code>rgba(96,165,250,0.12)</code>)
student accent <code>--sa-purple</code>	<code>#38BDF8</code> (soft <code>rgba(56,189,248,0.055)</code>)

Light mode

Token	Value
page <code>--ds-surface-page</code>	<code>#F7F9FC</code>
card	<code>#FBFCFF</code> · raised <code>#FDDEFF</code> · muted <code>#F3F6FA</code>
text primary/secondary/muted	<code>#0F172A</code> / <code>#475569</code> / <code>#64748B</code>
accent primary/secondary	<code>#2563EB</code> / <code>#7C3AED</code>
success/warning/error/info	<code>#2E7D32</code> / <code>#A16207</code> / <code>#B3261E</code> / <code>#0891B2</code>
border soft/medium/strong	<code>#E7EDF5</code> / <code>#CBD8E8</code> / <code>#94A3B8</code>

Dark elevation ladder (audit every screen against this): page `#0A0A0F` < recessed `#111117` < card `#16181F` < raised `#1C1F27`.

4.2 Card system

- **Dark:** gradient `linear-gradient(145deg, rgba(200,210,220,.075), rgba(180,195,210,.038) 50%, rgba(160,175,190,.016)) over rgba(28,30,38,0.86)`; border transparent; shadow `0 20px 46px -30px rgba(0,0,0,0.66)`; inset hairline `inset 0 1px 0 rgba(210,220,230,0.09)`; **glass** `blur(18px) saturate(1.35)`.
- **Light:** fill `linear-gradient(180deg, rgba(255,255,255,0.98), rgba(250,253,255,0.96))`; border `rgba(37,99,235,0.14)`; shadow `0 24px 70px -42px rgba(30,64,175,0.36)`; no blur (flat).
- **Dark shadow ladder:** xs/sm/md/lg as in tokens (`0 8/12/18/26px ...`).

4.3 Typography

- **Families:** display/body/brand = **Plus Jakarta Sans**; handwriting (note canvas only) = **Patrick Hand**; mono = platform mono.
- **⚠️ Fonts in repo** are `.woff2` (web-only). Flutter needs `.ttf` / `.otf`. **Action:** re-source the same families as TTF — Plus Jakarta Sans weights 400/500/600/700/800/900 and Patrick Hand 400 (both are the same families, open-licensed; Patrick Hand `.ttf` already exists in repo). Bundle under `mobile/assets/fonts/`. Note custom weight **450** (`--type-weight-regular`): map to a 400/500 face or use variable font.
- **Sizes:** 2xs 11 · xs 12 · sm 13 · md 14 · base 16 · lg 17 · xl 18 · 2xl 22 · 3xl 28→31 · 4xl 36→40.
- **Leading:** tight 1.08 · title 1.16 · snug 1.28 · normal 1.45 · body 1.58 · relaxed 1.72.
- **Text styles:** display 900/36 · page-title 800/28 · section-title 800/18 · card-title 800/17 · body 450/16 · body-strong 700/16 · ui 700/14 · label 800/12 · caption 600/13 · metric 800/28.

4.4 Radius / spacing / motion

- **Radius:** card 22 · card-compact 18 (mobile) · inner 14 · nav 14.
- **Spacing (4px grid):** 4/8/12/16/24/32. Page pad x 24 / y 16; section gap 24; card pad 24→16 mobile; control height / touch target 44. Responsive pad `clamp(16, 2.2vw, 24)`.
- **Motion easing:** ease-out `cubic-bezier(0.16,1,0.3,1)` · ease-in `cubic-bezier(0.4,0,1,1)` · standard `cubic-bezier(0.23,1,0.32,1)` → Dart `Cubic(...)` curves.
- **Durations:** micro 140 · hover 150 · dropdown 180 · modal 220 · route 240 · card-enter 220.
- **Route entrance:** `translateY(10px)→0`, ease-out 220ms.
- **Reduced motion:** if `MediaQuery.disableAnimations` (honors iOS Reduce Motion / Android animator scale) → snap all durations to ~1ms, exactly like the web.

4.5 Button hierarchy (design direction, 2026-06-13)

- **Hero CTA (rare, accent):** the **blue→indigo gradient** button (`linear-gradient(135deg, #3B82F6 → #6366F1)`), white text, **flat — no glow** is reserved for a **single** call-to-action in the **dashboard hero** — currently **"Start daily goal."** Use **at most one per screen**, effectively one in the app's main hub. It is an accent, **not** the default button.
- **Primary (everyday):** **solid flat brand blue**, no gradient — dark `#60A5FA` with dark text (`#04121F`); light `#2563EB` with white text. Used for Start exam, Save, Submit, etc.
- **Secondary:** neutral elevated (raised surface `#1C1F27` + hairline border).

- **Soft / tertiary:** primary-light tint (`rgba(96,165,250,0.12)`) + blue text (e.g. " New note").
- **Ghost / link:** text-only where appropriate. This hierarchy is encoded once in the Flutter button theme so the gradient never leaks onto ordinary actions. **All buttons are flat** — no glow, no drop shadow, light hairline borders only; press feedback is a subtle scale/opacity dip (iOS-style).

4.6 Native iOS feel — no 3D (design direction, 2026-06-13)

The app should feel like a native iOS app: clean, flat, content-first.

- **No 3D / skeuomorphic effects:** no card-flip rotation (`rotateY` /perspective), no tilt, no extruded depth. The flashcard uses a **flat cross-fade reveal** instead (§11.3).
- **No glows** anywhere (buttons, tab indicator, status dots, logo) and **no heavy strokes** — borders are light hairlines (`--line-soft`); icon strokes are thin, SF-symbol-like (~1.5–1.7 weight).
- **Depth comes only from** (a) iOS materials — the frosted-glass blur/translucency on cards (`UIVisualEffectView` / `BackdropFilter` , kept because it *is* native iOS), and (b) **subtle** ambient shadows — softened from the web values.
- Honors this on both platforms (the Flutter codebase is shared), but the *feel* target is iOS HIG; Android inherits the same flat language.

4.7 Native animations — everywhere (design direction, 2026-06-13)

Lean into rich, **native-feeling motion throughout** — but still **no 3D** (§4.6): everything is spring / translate / fade / scale, never rotation. Encode one shared motion kit so every screen feels alive:

- **Page transitions:** iOS push/pop (slide-from-right + interactive back-swipe, `CupertinoPageRoute`); modals as bottom **sheets** that spring up; tab switches cross-fade.
- **Shared-element (Hero) transitions:** course card → course detail, flashcard → expanded, lesson tile → reader, result row → result page.
- **Staggered list entrance:** rows fade + slide-up with ~30–40 ms stagger (`flutter_staggered_animations`).
- **Spring physics** on gestures: sheet drag-to-dismiss, pull-to-refresh, swipe actions on rows, card/button press (subtle scale + opacity dip).
- **Micro-interactions:** tab pill slide, bookmark/heart toggle, checkbox, segmented controls, switches — all animated.
- **Loading:** skeleton **shimmer** placeholders during boot/lazy loads (not spinners) → content cross-fades in.
- **Number / progress motion:** score-ring count-up, progress-bar fills, metric count-ups, streak pulse.
- **Onboarding:** parallax between welcome slides; mascot idle micro-motion; animated page dots.
- **Reduced motion:** all of the above collapse to ~1 ms / plain cross-fade when the OS setting is on. Flutter tooling: `AnimationController` / `AnimatedSwitcher` / implicit animations, `Hero` , `CupertinoPageRoute` , `SpringSimulation` , `RefreshIndicator` / `CupertinoSliverRefreshControl` , `shimmer` , `flutter_staggered_animations` .

5. Navigation & responsive shell

Replicates `AppShell` / `StudentPanelLayout` / `AppSidebar` .

5.1 Breakpoints

`<380` (tight phone, hide tab labels) · `≤900` (phone/tablet portrait → bottom tab bar) · `≥901` (desktop/large iPad landscape → left sidebar). Secondary content breakpoints 430 / 601 mirrored where the web uses them.

5.2 Mobile (≤900): bottom tab bar

- **5 tabs:** Courses · Q-Bank · **Study Hub** (center) · Study · Results — iOS-18-style floating pill indicator; 64px height + `safe-area-inset-bottom` .
- The **"Study"** tab is a grouped hub that routes to Planner / Flashcards / Lessons (AI Notes) / Saved (Bookmarks) / Notes (matchPaths behavior from the web shell).
- `<380px` : hide tab labels, shrink icons.
- Focus-mode routes (taking a quiz/exam, AI-note reader, review) **hide the tab bar** (full-screen), same as web.

5.3 Desktop / large tablet (≥901): sidebar

- Left drawer 280px, collapsible to 80px (220ms animation). 10 items in order: **Study Hub, Courses, Planner, Lessons, Flashcards, Q-Bank, Exams, Results, Saved, Subscriptions.**
- Content offset right; active item = brand-primary color + glow shadow; sidebar footer = avatar initials + "Medical Student".
- Cmd/Ctrl-K search parity → in-app search sheet.

5.4 Safe area / system insets

Use `SafeArea` + `MediaQuery.padding` to replicate `--lms-safe-*`. iOS notch / Dynamic Island and Android status/nav bars handled natively (no CSS `env()` needed). Content uses full dynamic height (Flutter handles `100dvh` equivalent automatically).

5.5 First-run & launch flow (welcome → login → create account)

1. **First-ever launch:** native splash → `/welcome` onboarding (2–3 parallax slides introducing Q-Bank/exams, AI notes/flashcards, progress/planner) → `/auth/login`.
2. **Later launches:** splash → if a stored session validates via `GET /auth/me` → `/app/dashboard`; otherwise → `/auth/login`. Onboarding is shown **only once**, gated by a persisted `seenOnboarding` flag (`shared_preferences`).
3. **Create account:** from login, "Create account" → `/auth/register` (`POST /auth/register`, **same backend + MySQL DB**) → dashboard (Free plan; push notifications default **off** until opt-in). All auth, accounts, and data come from the **existing NestJS API + MySQL** — the native app keeps no server/data of its own.

6. Screen build list (all student screens)

Each screen = one Flutter route (go_router) ported from the named web component, wired to the listed endpoints, reproducing the listed effects. Public `/auth/*` and authenticated `/app/*`.

Auth (public)

Native route	Web source	Endpoints	Key effects
<code>/welcome</code>	(new — native onboarding)	none (local, <code>seenOnboarding</code> flag)	First-run only: 2–3 parallax welcome slides + animated dots → "Get started" → login. Skipped on later launches (§5.5)
<code>/auth/login</code>	LoginPage	<code>POST /auth/login</code> , <code>POST /auth/google</code> (+ <code>/auth/google/code</code>)	Animated 4-layer radial-gradient + grid-mask background; Google sign-in
<code>/auth/register</code>	RegisterPage	<code>POST /auth/register</code>	Create account — full name/email/password/confirm/terms; password ≥10; → Free plan; "account created"; same backend + DB
<code>/auth/forgot-password</code>	ForgotPasswordPage	<code>POST /auth/forgot-password</code>	—
<code>/auth/reset-password</code>	ResetPasswordPage	<code>POST /auth/reset-password</code>	Token from deep link

Student app (authenticated)

Native route	Web source	Endpoints	Key effects / notes
<code>/app/dashboard</code>	StudentDashboardPage	GET <code>/student/boot</code> (then <code>/student/dashboard</code>), POST <code>/student/dashboard/activity</code>	Page hero + metric chips; study mascot ; progress, weak topics, daily goals
<code>/app/pending</code>	DashboardPage (pending)	GET <code>/auth/me</code>	Awaiting-approval holding state
<code>/app/profile</code>	ProfilePage	PATCH <code>/auth/profile</code> , PATCH <code>/auth/password</code>	Avatar picker (procedural, §13), password mgmt
<code>/app/courses</code>	StudentCoursesPage	GET <code>/student/courses</code>	Course library, lesson progress; subject color palettes
<code>/app/courses/:id</code>	CourseDetailPage	GET <code>/student/courses/:id</code> , PATCH <code>/student/courses/lessons/:lessonId/progress</code>	<code>csum=*</code> flat summary; dynamic subject color (cardiology=rose, etc.)
<code>/app/quizzes</code>	StudentQuizzesPage (practice)	GET <code>/student/quiz-attempts/quizzes</code>	Q-Bank practice library
<code>/app/exams</code>	StudentQuizzesPage (exam)	GET <code>/student/quiz-attempts/quizzes</code>	Exam library (timed)
<code>/app/quizzes/:quizId</code>	TakeQuizPage	GET <code>.../quiz/:quizId</code> ; practice: <code>.../practice/:quizId/save draft finish</code> , <code>.../answer/:qid/prewarm reveal</code> ; exam: <code>.../exam/:quizId/save submit</code> ; POST <code>/student/question-reports</code>	Focus mode (no tab bar); timer; haptics; report question
<code>/app/quizzes/:quizId/practice-review</code>	PracticeReviewPage	GET <code>.../practice-review/:quizId</code>	Answers + explanations
<code>/app/results</code>	ResultsListPage	GET <code>.../results</code>	History list, filter/sort
<code>/app/results/:attemptId</code>	ResultPage	GET <code>.../result/:attemptId</code>	Animated conic-gradient score ring + count-up
<code>/app/review/:attemptId</code>	ReviewPage	GET <code>.../review/:attemptId</code> , POST <code>.../review/:attemptId/complete</code>	Full incorrect-answer review workspace
<code>/app/ai-notes</code>	AiNotesListPage	GET <code>/student/ai-notes</code>	Lessons/notes library, search
<code>/app/ai-notes/:id</code>	AiNotesPage	GET <code>/student/ai-notes/:id</code>	Note canvas + procedural pen/scribble

Native route	Web source	Endpoints	Key effects / notes
			sound (§11.4), paper texture
/app/study/lesson/:lessonId	AiNotesPage (lesson)	GET /student/ai-notes/lesson/:lessonId , GET /student/lessons/:id , lesson annotations CRUD .../lessons/:lessonId/annotations	Focused canvas edit; highlights/notes
/app/flashcards	StudentFlashcardsPage	(deck source via study endpoints)	Flip animation ; 11 card types color-coded
/app/notes	StudentNotesPage	lessons + annotations endpoints	Personal notes + lesson reader
/app/planner	StudyPlannerPage	GET /student/planner , GET /student/planner/agenda , POST /student/planner , PATCH/DELETE /student/planner/:id	Agenda, tasks, due dates, reminders
/app/study	StudentStudyPage	—	Hub → AI Notes / Flashcards / Planner
/app/bookmarks	BookmarksPage	GET /student/bookmarks , POST /student/bookmarks/toggle	Saved items, filter by type
/app/notifications	StudentNotificationsPage	GET /student/notifications , POST /student/notifications/:id/read	Inbox; mark read
/app/subscriptions	StudentBillingPage	GET /plans , GET /subscriptions/me , POST /subscriptions/request , .../payhere/initiate , .../manual-payment/request , PATCH /subscriptions/:id/extend renew cancel payment , DELETE /subscriptions/requests/:id	Plans, status, payment history
/app/subscriptions/checkout/:planId	StudentCheckoutPage	POST /subscriptions/payhere/initiate → external browser (§14)	PayHere; return via deep link
(settings)	settings.api	GET /settings/public , .../availability , POST /settings/availability/unlock	Branding/feature flags, availability gating

7. Backend / API integration

The backend is **unchanged**. The Flutter app calls the same endpoints the web app calls. Authoritative endpoint list is the frontend api wrappers in `frontend/src/shared/api/*.api.js` (13 modules). Full table in §6; the complete set is enumerated and grouped by feature (auth, profile, boot, dashboard, courses, quizzes/exams, results/review, ai-notes, bookmarks, notifications, planner, push, subscriptions, settings).

7.1 Base URL (`config.dart` , via `--dart-define`)

The web resolves base URL per platform (`frontend/src/shared/platform/config.js`). Native equivalent:

- **Dev (simulator/emulator):** iOS sim → `http://localhost:3000/api` ; Android emulator → `http://10.0.2.2:3000/api` ; physical device → `http://<LAN-ip>:3000/api` .
- **Prod:** an **absolute** HTTPS origin (web's same-origin `/api` doesn't exist for a native app). 🚩 **Open item:** you must provide the production API origin (see §18).

7.2 Dio client parity with Axios (`frontend/src/shared/api/client.js`)

- Inject `Authorization: Bearer <token>` when a token exists.
- Timeout: **10s native** default (web uses 30s; native config uses 10s) — configurable.
- **Retry** with exponential backoff (default 2 retries; retry writes), via `dio_smart_retry` + custom logic; base-URL fallback list on network failure.
- On **401** → clear auth, route to login, optional session-lock overlay if user was authenticated.
- **Redaction**: strip tokens/passwords/auth codes from all logs (parity with web).
- Light **response cache** for read endpoints (mirror `createTimedApiCache`).

7.3 Boot batching

On launch call `GET /student/boot` — one request returning 6 slices (dashboard, notifications, planner/agenda, quizzes, bookmarks, ai-notes) via `Promise.allSettled` server-side; null slices fall back to standalone endpoints. The Flutter app hydrates the dashboard/study-hub from boot, then lazy-loads detail screens.

Path note: frontend wrappers call `/student/boot`, `/student/dashboard`, etc. The backend module scan reported `/boot/student`, `/dashboard/student`. These must be reconciled against the live NestJS controllers at kickoff (1-hour verification task) — the wrappers are the source of truth for what the app sends.

8. Auth & session

Mirrors `authStore.js` + `authToken.js` (native path).

- **Token model:** server returns `sessionToken` on login/register/google. Store it in **flutter_secure_storage** (Keychain / EncryptedSharedPreferences) under a `lms_native_session_token` key, plus cached `user`.
- **Header:** every request sends `Authorization: Bearer <token>` (no httpOnly cookie on native — that's the web path).
- **Hydration:** at app boot, probe `GET /auth/me` (short timeout); set `isAuthenticated` / `user`; show splash until resolved. `go_router` redirect guard sends unauthenticated users to `/auth/login`.
- **State keys (Riverpod):** `token`, `user`, `isAuthenticated`, `isHydrating`, `authNotice`, `sessionExpiredLock` — same shape as Zustand store.
- **Sign-up:** `POST /auth/register {fullName,email,password,confirmPassword,acceptedTerms}` → set token+user → land on dashboard (Free plan). New-signup notification-off default replicated (push prefs default off until user opts in).
- **Google sign-in:** `google_sign_in` → POST credential to `/auth/google` (or code to `/auth/google/code`). ⚠ Needs iOS + Android OAuth client IDs (see §18).
- **Logout:** `POST /auth/logout` + clear secure storage + unregister push token (`DELETE /push/native-token`).

9. Native platform features (replace Capacitor plugins)

Current Capacitor config: App ID `com.erpm.medical.lms` (the **new native apps use a rebranded ID** — `app.xyndrome.lk`), name **xyndrome**, status bar/splash `#0b121b`.

Capability	Capacitor (current)	Flutter native plan
Push	@capacitor/push-notifications (APNs/FCM)	<code>firebase_messaging</code> ; register device token via <code>POST /push/native-token</code> , unregister on logout. iOS: APNs entitlement + Firebase. Android: FCM.
Local notifications	@capacitor/local-notifications; channels <code>default</code> , <code>exam_reminders</code> , <code>course_updates</code> , <code>account_alerts</code> (vibration+lights)	<code>flutter_local_notifications</code> ; recreate the 4 Android channels exactly ; iOS categories.
Haptics	@capacitor/haptics (impact / selection / transient / notification success-warning-error)	<code>HapticFeedback</code> + richer set via platform channel; map each web call site to the matching generator. Honor system haptic/reduced-motion settings.
Status bar	dark overlay <code>#0b121b</code>	<code>SystemChrome.setSystemUIOverlayStyle</code> matching <code>#0b121b</code> .
Splash	auto-hide, <code>#0b121b</code>	<code>flutter_native_splash</code> bg <code>#0b121b</code> , logo mark.
Keyboard	@capacitor/keyboard overlay + listeners	Native insets / <code>MediaQuery.viewInsets</code> ; scroll-into-view for focused fields.
App lifecycle	@capacitor/app	<code>WidgetsBindingObserver</code> / <code>AppLifecycleState</code> .
Deep links	(PWA shortcuts)	<code>app_links</code> / Universal Links + App Links for: password reset, PayHere return, push taps, and shortcuts (Courses / Quizzes / Lessons).

9.4 Open server items (small, additive — confirm at kickoff)

- `POST /push/native-token` already exists ✅ — confirm payload shape (platform, token).
- PayHere return URL must redirect to a deep link the app can catch (or app polls `GET /subscriptions/me`). May need one config value server-side. Confirm.

10. Responsive / multi-form-factor

The web is mobile-first with phone/tablet/desktop layouts; Flutter replicates with `LayoutBuilder` + `MediaQuery` :

- Phone (≤ 900):** single column, bottom tab bar, compact card radius 18, page pad 16–24.
- Tablet / iPad (601–900 portrait, ≥ 901 landscape):** wider content column (mirror the web's single forced width ~1080 where applicable), sidebar appears ≥ 901 .
- Desktop (≥ 901):** sidebar 280/80, multi-column where the web uses it.
- Test matrix in §17 covers iPhone SE → Pro Max, iPad mini → 12.9", common Android phones + tablets, and large-window desktop.

11. The hard parts (effects) — replication strategy

11.1 Dark glassmorphism cards

`BackdropFilter(ImageFilter.blur(18,18))` inside a `ClipRRect(22)` , layered with: the 3-stop diagonal grey gradient over `rgba(28,30,38,0.86)` , an inner 1px hairline (`inset 0 1px 0 rgba(210,220,230,0.09)` via a top border-gradient), and the ambient shadow. The `saturate(1.35)` needs a **color-matrix shader** (`ColorFilter.matrix`) layered with the blur, since `ImageFilter.blur` alone won't saturate. Build one reusable `GlassCard` widget; tune blur/saturation per platform if needed. Light mode = flat (no backdrop filter).

11.2 Score ring (Result page)

`CustomPainter` drawing a conic/sweep gradient arc + `AnimationController` count-up of the number, ease-out 220–600ms; reduced-motion → instant.

11.3 Flashcard reveal + 11 card types

No 3D flip (per §4.6). Tapping a card does a **flat cross-fade reveal** (front fades/slides out, answer fades/slides in — `AnimatedSwitcher` / opacity + small `translateY`), the native-iOS feel. Color-coded badges per the 11 types (Q&A, Definition, Mechanism, Features, Management, Classification, Causes, Diagnosis, Complications, Mnemonic, Key Points, Summary).

11.4 Note canvas + procedural scribble/drawing audio (highest-risk item)

The web generates audio **on the fly with the Web Audio API** — **there are no sound files** (`NoteCanvas.jsx` lines 612–698; `AiNotesPage.jsx` `startDrawingStrokeSound` / `modulate` / `stop`). Two behaviors to replicate:

1. **Scribble (typing)**: 55ms filtered-noise burst, bandpass 1700–3000 Hz $Q \approx 6$, throttled ≥ 58 ms (~ 17 /s), fired on text input.
2. **Drawing ('spen')**: 3 looped procedural buffers (main/texture/vertical) modulated in real time by pen **pressure, speed, tilt, direction**; throttle 95–320ms. Plus a 'secret' mode (fart/dog/cat/boing/squeak).

Audio — Flutter approach: use `flutter_so_loud` (low-latency engine that supports raw PCM buffers + real-time volume/pitch modulation and looping). Either (a) synthesize equivalent noise buffers at runtime, or (b) since the scribble is deterministic, **pre-render a few short PCM/wav clips** offline and modulate playback rate/volume per stroke (simplest, near-identical result). For Apple Pencil / S-Pen, read `PointerEvent.pressure`, `tilt`, and compute speed/direction from successive points. Mark this as a focused spike (1–2 days) with a fidelity check against the web.

Ink / drawing surface — two fidelity tiers:

- **Tier A (default, one codebase)**: `CustomPaint` + `perfect_freehand` (Dart port) for natural pressure-aware strokes, rendered on **Impeller** (Flutter's Metal-backed iOS renderer); paper grain/tooth via a tiled texture or noise shader; handwriting text uses **Patrick Hand**. Smooth at 120 Hz; perceived latency is ~ 90 –95% of Apple Notes — indistinguishable for the vast majority of note-taking.
- **Tier B (equal-to-native ink, recommended upgrade)**: embed the platform's **real ink engine as a native view** — `UIKitView` hosting `PKCanvasView` (`PencilKit`) on iOS and `AndroidView` hosting a native ink/ `MotionEvent` canvas on Android — while the rest of the AI-notes screen (lists, reader, callouts, offline-first store, sync) stays shared Flutter. This gives Apple's predictive-touch, ~ 9 ms ink — **literally equal to a fully-native Swift app** — at the cost of a small per-platform native module to maintain. Choose Tier B if Apple-Notes-grade handwriting is a priority; otherwise ship Tier A and upgrade the canvas later without touching the rest of the app.

11.5 Animated login background

4-layer radial-gradient (cyan + indigo + blue + sky) + grid-mask, slowly animated. Flutter: stacked `RadialGradient` containers + a `CustomPainter` grid with a mask, animated via a long-running `AnimationController`; reduced-motion → static.

11.6 Haptic vocabulary

Map every web haptic call site (impact / selection / transient / success-warning-error) to the matching Flutter generator so the feel matches; honor system settings.

11.7 Reduced motion

Global: read `MediaQuery.disableAnimations`; a `motion.dart` helper returns ~ 1 ms durations when true — applied to all of the above.

12. Branding & assets pipeline

Reuse the **exact** brand assets so it matches pixel-for-pixel.

Asset	Source	Native use
Logo mark (gradient X + stethoscope)	<code>frontend/public/brand/xyndrome-logo-mark-{light,dark}.webp</code>	In-app logo. Files named by background: dark theme → <code>-dark.webp</code> , light theme → <code>-light.webp</code> (per <code>LandingNav.jsx</code>). Convert webp→png/svg for crisp native scaling.
App icon source	<code>pwa-icon.svg</code> , <code>pwa-icon-512.png</code> , <code>apple-touch-icon.png</code>	Generate iOS icon set + Android adaptive icon (<code>pwa-maskable*</code> as adaptive foreground) via <code>flutter_launcher_icons</code> .
Fonts	Plus Jakarta Sans (<code>.woff2</code>) + Patrick Hand (<code>.ttf</code>)	Re-source PJS as <code>.ttf</code> (same family); bundle both under <code>assets/fonts/</code> .
Medical icons	<code>frontend/public/medical/*.svg</code> (10: blood, brain, dna, heart, lungs, microscope, pills, stethoscope, syringe, tooth)	<code>flutter_svg</code> .
Course/quiz art	<code>frontend/public/lms-assets/quiz-course-art/{maternal,medicine,surgery,systems}-cutout.{png,webp}</code>	Bundle as image assets.
Mascot	<code>StudyMascot.jsx</code> (5 inline-SVG variants: stetho, brain, readiness, streak, review) + generated webp mascots in <code>public/temp/mascots/</code>	Port SVG variants to <code>flutter_svg</code> widgets (preferred) or bundle webp.
Avatars	<code>profileAvatarData.js</code> (6 procedural variants: skin/hair/shirt combos)	Port the generator to a Dart <code>CustomPainter</code> /SVG.
UI icons	inline JSX SVG (<code>ActionIcons.jsx</code> , nav icons)	Port to a Dart icon set (SVG or <code>IconData</code>).
Manifest meta	<code>manifest.webmanifest</code> : name "xyndrome", theme/bg <code>#05070d</code> , shortcuts Courses/Quizzes/Lessons	Drives app name, splash color, home-screen quick actions.

13. App identity & store metadata

- **Name:** xyndrome. **Bundle / Application ID:** `app.xyndrome.lk` (rebranded from the old `com.erpm.medical.lms`; `erpm` dropped). Because the ID changed, these are **new App Store / Play listings**, not an in-place update of the old `erpm` app. Flutter package name = `xyndrome`; the iOS `PRODUCT_BUNDLE_IDENTIFIER` and Android `applicationId` / `namespace` are set to `app.xyndrome.lk`. Use a temp dev id `app.xyndrome.lk.dev` during development to install side-by-side with any existing build.
- Categories: Education / Medical (from manifest).
- iOS: APNs, Sign in with Google, Universal Links entitlements. Android: FCM, App Links, adaptive icon.

14. Billing (your choice: external web checkout) — with risk flagged

- **Plan:** the native Subscriptions screen reads plans + status (`GET /plans`, `GET /subscriptions/me`). Checkout opens the existing **PayHere** web flow (`POST /subscriptions/payhere/initiate` → payment URL) in the **system browser** via `url_launcher`; on completion the user returns via deep link and the app refreshes `GET /subscriptions/me`. Manual-payment and request flows reuse the same endpoints.
- **⚠ Store-policy risk (on record):** Apple App Store (Guideline 3.1.1) and Google Play generally **require their in-app purchase systems for digital subscriptions** and reject apps that route digital-goods purchases to an external web checkout. This can block approval. Mitigations if rejected: (a) hide purchase in the app and present plans as informational only, or (b) implement StoreKit / Play Billing later. We build external-checkout for v1 as decided and keep IAP as a fast-follow option.

15. Configuration & environments

- `--dart-define` flavors: `dev`, `staging`, `prod` → API base URL, Google client IDs, Firebase config, PayHere return URL.
 - `mobile/lib/config/config.dart` reads these; no secrets committed.
 - Firebase project(s) for FCM/APNs; `google-services.json` (Android) + `GoogleService-Info.plist` (iOS).
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16. Tooling, build & release

- **Toolchain:** Flutter stable, Dart 3, Xcode (iOS), Android Studio / Gradle (Android).
 - **CI/CD (proposed):** GitHub Actions / Codemagic — analyze + test on PR; build IPA/AAB on tag; deliver to **TestFlight + Play Internal testing**.
 - **Signing:** iOS provisioning + APNs key; Android keystore (Play App Signing).
 - **Versioning:** semver + build number; match store track.
-

17. Testing & QA

- **Unit/widget tests:** theme tokens, API client (auth header, 401, retry, redaction), repositories, key widgets (GlassCard, ScoreRing, Flashcard).
 - **Golden tests:** snapshot critical screens in light + dark to lock visual fidelity.
 - **Integration:** auth flow, boot hydration, take-quiz (practice + exam), result, note canvas.
 - **Responsive matrix:** iPhone SE/14/15 Pro Max; iPad mini/Air/12.9"; Pixel/Samsung phones + tablet; large desktop window. Verify bottom-tab ↔ sidebar switch at 900/901, safe-area on notch/Dynamic Island, dark elevation ladder on every screen.
 - **Fidelity audit:** side-by-side vs. web for each screen (color tone, radius, shadows, motion timing, glass).
 - **Accessibility:** reduced-motion, dynamic type, contrast (Apple HIG semantics), 44px touch targets.
-

18. Risks & open items (need from you / to confirm at kickoff)

1. **Production API origin (HTTPS)** — required; native can't use same-origin `/api`. *(blocking for prod build, not for dev)*
 2. **Google OAuth client IDs** (iOS + Android) for `/auth/google`. *(blocking for Google sign-in)*
 3. **Firebase / APNs setup** — Firebase project + APNs auth key for push. *(blocking for push)*
 4. **PayHere return deep link** — confirm return URL → app deep link (or poll `subscriptions/me`).
 5. **Billing store policy** — accept rejection risk of external checkout, or plan IAP fast-follow (§14).
 6. **Procedural audio fidelity** — confirm the soloud / pre-rendered approach matches the web's pen feel (spike).
 7. **API path reconciliation** — `/student/boot` (frontend) vs `/boot/student` (backend scan) — 1h verification.
 8. **Bundle ID** — ✅ decided: `app.xyndrome.lk` (new listings; rebrand from `com.erpm.medical.lms`).
 9. **Fonts** — confirm re-sourcing Plus Jakarta Sans as `.ttf` (same family, open license).
 10. **"Free stuff" definition** (offline-first scope, §3.3) — confirm it means the student's own notes/annotations **plus** backend-flagged free (non-premium) lessons/flashcards/quizzes. Does the API expose a `free` /premium flag per item?
-

19. Work breakdown (ordered tasks — "one task at a time," built into one app)

All screens ship together (your choice), but built in this dependency order so each task is runnable/verifiable.

- Create `mobile/` Flutter app; configure flavors (`dev/staging/prod`), bundle IDs, splash, launcher icons, fonts.

T1 — Design system

- `tokens.dart` (all \$4 values) + light/dark `ThemeData` ; `GlassCard` , shadows, typography, motion helper, reduced-motion.

T2 — Networking & auth core

- Dio client (auth header, timeout, retry, 401, redaction, base-URL fallback); secure storage; Riverpod auth store; `/auth/me` hydration; `go_router` with auth redirect.

T3 — Navigation shell

- Responsive shell: bottom tab bar (5) ≤900 + sidebar (10) ≥901; focus-mode hiding; safe areas; page hero + metric chips.

T4 — Onboarding + Auth screens

- First-run **welcome/onboarding** (2–3 parallax slides + animated dots, persisted `seenOnboarding`) → login (§5.5).
- Login (animated bg + Google), **Create account / Register** (`POST /auth/register` , same DB), Forgot, Reset (deep link).
- Shared **motion kit** (§4.7) wired here first: page transitions, hero, staggered entrance, shimmer.

T5 — Dashboard / Study Hub

- Boot batching hydration; metrics, weak topics, daily goals, mascot.

T6 — Courses

- Library + course detail (subject color palettes, lesson progress).

T7 — Q-Bank / Exams + Take Quiz

- Practice + exam libraries; TakeQuiz focus mode (timer, save/draft/finish, reveal/prewarm, exam submit, report question); haptics.

T8 — Results & Review

- Results list; Result page (score ring + count-up); Review workspace; practice-review.

T9 — AI Notes / Lessons + Note Canvas

- List + reader; **note canvas** (CustomPaint + pressure) + **procedural audio spike**; annotations CRUD; paper texture; Patrick Hand.

T10 — Flashcards

- Flip animation + 11 color-coded types.

T11 — Planner / Notes / Bookmarks (+ offline-first slice, §3.3)

- Planner agenda CRUD; personal notes; bookmarks toggle/filter.
- Local DB (Hive/Drift) as source of truth for **personal notes** + **free content**; offline create/edit; background sync queue (dirty flags, last-write-wins); offline/"saved on device" indicators.

T12 — Notifications + Push

- Inbox + mark-read; FCM/APNs registration to `/push/native-token` ; 4 local-notification channels; push prefs (default off for new accounts).

T13 — Profile

- Profile edit, avatar picker (procedural), password change.

T14 — Subscriptions + external checkout

- Plans + status; PayHere external browser checkout + deep-link return.

T15 — Settings / availability gating

- `/settings/public` + availability unlock.

T16 — Polish & fidelity pass

- Side-by-side audit vs web; glass tuning; motion timing; reduced motion; accessibility; golden tests.

T17 — Release engineering

- CI/CD, signing, TestFlight + Play internal; store assets.
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20. Definition of done

- Both apps build and run natively on iOS + Android (no webview).
 - Every student screen in §6 present, wired to the same backend, visually matching the web (light + dark) within the fidelity audit tolerance.
 - Bottom-tab ↔ sidebar responsive behavior correct across the test matrix; safe areas correct.
 - Auth (email + Google), boot hydration, take-quiz, results, note canvas + scribble feel, flashcards, planner, notifications + push, subscriptions external checkout all functional.
 - Reduced-motion, haptics, and dark elevation ladder verified.
 - TestFlight + Play internal builds delivered.
-

Appendix A — Source-of-truth files in the existing app

- Design tokens: `frontend/src/shared/styles/00-tokens/` + `01-base/theme.css`
- Nav shell: `frontend/src/shared/layout/{AppShell,StudentPanelLayout,AppSidebar}.jsx`
- Auth: `frontend/src/shared/stores/authStore.js` , `authToken.js`
- API client: `frontend/src/shared/api/client.js` ; wrappers: `frontend/src/shared/api/*.api.js`
- Platform/base-URL: `frontend/src/shared/platform/config.js`
- Note canvas + audio: `frontend/src/surfaces/app/student/ai-notes/{NoteCanvas,AiNotesPage}.jsx`
- Capacitor config: `frontend/capacitor.config.ts`
- Assets: `frontend/public/brand/` , `frontend/public/medical/` , `frontend/public/lms-assets/` , `frontend/src/shared/assets/fonts/`
- Manifest: `frontend/public/manifest.webmanifest`